



# MURRAY LEWIN

CONSULTANT ENGINEER

## CONTACT DETAILS

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## EDUCATION

**Bachelor of Science, University of Western Australia**  
Mechanical Engineering  
Data Science

## TECHNICAL SKILLS

- SAP
- Power BI
- Power Apps
- SQL
- Python
- MATLAB

## PERSONAL SKILLS

- Stakeholder Engagement
- Collaboration
- Process Improvement
- Clear Communication
- Analytical Thinking
- Continuous Learning
- Adaptability

## ASC - ASSET ENGINEERING

Dear ASC Recruitment Team,

I am applying for an asset engineering role with ASC. At Deloitte, I work with maintenance data, reliability analysis and engineering applications. I am passionate about working with data, improvement and optimisation, and creating useful dashboards, applications and insights for engineering teams. I would welcome the opportunity to contribute these skills to ASC's submarine sustainment work and develop my experience in defence asset engineering.

Kind regards,

**Murray Lewin**

## WORK EXPERIENCE

**Deloitte Touche Tohmatsu Limited Australia**

**Jan 2024 – Present**

**See Project Activities Below**

**Resources Sector Client**

***Maintenance System Optimisation (MSO), Consultant Engineer***

*January 2026 - Current*

Deloitte delivered a full-stack, cloud-native AI application for the client to modernise and automate maintenance strategy processes across Heavy Mobile Equipment (HME) and Fixed Plant (FXP) assets. The platform replaces fragmented, manual engineering workflows with AI-augmented data review, statistical reliability modelling, and data-driven maintenance optimisation, enabling engineers to shift from reactive to proactive asset management. The solution integrates with Databricks and SAP work order systems to provide end-to-end traceability from raw operational data through to validated maintenance recommendations, delivering projected annual savings of \$35 million through reduced unplanned downtime and optimised component replacement cycles.

- Reliability engineering - Weibull analysis, MTBF/MTTF calculation, Asset Health Index scoring, component age analysis and replacement projection modelling.
- AI-augmented engineering workflows - human-in-the-loop review for AI-generated material classifications, FLOC BOM validation and work order classification.
- Full-stack web development - React/TypeScript (frontend) and Python/FastAPI (backend) following a vertical-slice architecture.
- Data engineering - Databricks Gold-layer enterprise data products and SAP work order data into a dual-schema PostgreSQL application database.
- Azure cloud platform - Azure Container Apps with Entra ID authentication and Application Insights telemetry.
- Data visualisation - interactive reliability charts including Weibull curves, age analysis timelines and health index dashboards using Plotly.js.

ACHIEVEMENTS

- Duke of Edinburgh's Award - Bronze

VOLUNTEERING

Ronald McDonald House – Big Brother Program (2018)

Supported children from rural and remote communities during hospital stays.

Ngalangangpum School – Warmun Community

Supported students and classroom activities in a remote Indigenous community.

Mt Hawthorn Special Education Unit

Assisted students with learning and physical disabilities.

Quintilian School

Supported classroom activities, sport and the school buddy programme.

Sienna Centre – Holy Rosary Primary School

Assisted classroom and social activities for children with disabilities.

REFEREES

Available on request

Continued Project Experience – Deloitte Touche Tohmatsu Limited

Global Resources Client

**Client Asset Management Framework (Phase 1 and 2), Consultant Engineer**  
*September 2025 – January 2026*

Client Asset Management Framework

Rollled out the Client Asset Management Framework and System Definition. Defined Key Practice Areas, system requirements and documentation layout for the client. Worked with the the client's central team across Australia, Chile, South Africa, Ghana & Canada to identify and align global frameworks.

- Asset Management Framework definition and Key Practice Areas rollout.
- Asset Management System documentation and global standardisation.
- International workshopping facilitated across 5 countries using Mentimeter.

Resources Sector Client

**Asset Health Index - Pilbara Sites, Consultant Engineer**  
*March 2025 – September 2025*

Asset Health Index – Two Pilbara Sites

4/3 roster fly-in fly-out rollout and build of the Asset Health Index for the client. This involved the wet plant assets at one Pilbara site and the low-grade fixed plant at a second Pilbara site.

- Power BI dashboard development to track and communicate asset health scores across sites.
- Engagement with Reliability Engineers and site stakeholders to validate AHI scoring methodologies.
- People engagement and RE collaboration to align the Asset Health Index with site maintenance strategies.

Resources Sector Client

**Labour Benchmarking Review, Data & Reporting Analyst**  
*January 2025 – March 2025*

Labour Benchmarking Review

- Worked with national resources to collect a database of information in order to draw conclusions and summaries on industry standard for labour quantities.
- Direct back-and-forth with client to ensure scope aligned with the desired outcomes and the correct conclusions were made.
- Breaking down each site into major assets using exports from various sources, allowing for accurate comparisons between equivalent datasets (ensuring bias due to resource type/org structure would not affect the end result).

Resources Sector Client

**Asset Management Plans (Substations), Engineering Analyst**  
*September 2024 – December 2024*

Asset Management Plans – Pilbara Site Substations

- Report Building and collation of large data, utilising appendices and other tools to improve reading ease.
- Substation site validation – Checking the correct regulatory precautions are taking in Low Voltage Substations at the site.
- Recommendations are made based on historical costing and Changeout breakdown of assets - ensuring that the best measures are taken to break down each substation in scope.

## **Continued Project Experience – Deloitte Touche Tohmatsu Limited**

### **Resources Sector Client**

#### ***Asset Management Uplift Programme, Graduate Team Member***

*June 2024 – September 2024*

Asset Tactics Review - Pilbara Site

- Bad Actor Review involving calculating the worst performing site assets on the basis of downtime and their associated criticality.
- Changeout Analysis of SAP exports involving review of PM01, PM02 and PM03 changeout notifications for incomplete changeouts to find causes of unnecessary downtime on assets. This also included looking at timing issues of PM01 and PM02 notifications.
- Documentation/PRT review to find gaps in task descriptions, or missing information which would make it challenging for an inspector to evaluate an assets condition.

### **Previous Experience**

#### **Macmahon (Mining Client)**

#### ***Closed Work Order and Parts Reallocation, Storefront***

*November 2022 – March 2024 – Part time /Ad Hoc*

Using SAP (S4Hana) to re-issue stock items back into the storefront that weren't fitted from maintenance orders. Also preparing work orders for fitters and maintenance crew to perform routine fixes.

- Associating parts with SAP ID so that they can be re-issued into stock.
- Collecting work orders and ordering parts to make service kits for the team.
- Operating a stock take which involved reviewing item quantities and adjusting SAP as such.
- Redistribution of hose stockpiles back into stock bins and disposing of out of date hoses to clear backlog.

### **Prior Experience – University of Western Australia**

#### **University of Western Australia**

#### ***Chatbot UWA Web Design, Group Member***

*July 2023 – November 2023*

Create a UWA questionnaire inspired chatbot that uses the ChatGPT algorithm to answer students questions from a webpage.

- HTML and CSS design by first building a framework with HTML and then adding colour and design with CSS.
- JavaScript actioning which involved strategically adding buttons and capability to the page such as saving user details, passwords and navigation.
- Python AI training was used to create a link between the AI from Amazon cloud servers and the various documents stored in the UWA questionnaire database. A trial and error system was used to bring the AI to ~95% accuracy.
- Final Presentation was made and the product was presented to the UWA Software Engineering board which resulted in an overall positive reaction, and 90% overall project mark.